

CLOUDFLARE RADAR

# 2024 Year in Review

Global Internet Traffic, Trends & Security Insights

Data observed through Cloudflare's global network in 2024

# 17% Global Traffic Growth and Starlink's 3.3x Surge

## GLOBAL INTERNET TRAFFIC

**+17%**

Year-over-Year Growth

Continuing the upward trend driven by increasing reliance on Internet for communication, commerce, and entertainment

## SPACEX STARLINK

**3.3x**

Traffic Surge in 2024

Satellite connectivity expanding to previously underserved regions worldwide

Key Insight: The trend line normalizes from mid-January to account for post-holiday activity patterns.

# Google Leads Top 10 Services as Generative AI Rises

| RANK | SERVICE   |
|------|-----------|
| #1   | Google    |
| #2   | Facebook  |
| #3   | Apple     |
| #4   | TikTok    |
| #5   | AWS       |
| #6   | Microsoft |
| #7   | Instagram |
| #8   | YouTube   |
| #9   | Amazon    |
| #10  | WhatsApp  |

Methodology: Rankings based on aggregate data from Cloudflare's 1.1.1.1 DNS resolver

## GENERATIVE AI TREND

### Meteoric Rise

Generative AI continued its rapid rise throughout 2024, fundamentally changing how users interact with Internet services.

Meanwhile, the Metaverse receded into the background, with cloud and AI infrastructure becoming the dominant narrative.

# Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

## POST-QUANTUM

2%→13%

TLS 1.3 Traffic

Chrome 124 default enabled April 2024

## HTTP/3

21%

Traffic Adoption

QUIC-based, encrypted by default

## GO LANGUAGE

12%

API Client Share

Most popular API client language

## Additional Technology Metrics

- **Mobile OS:** iOS accounts for 33% of traffic; Android exceeds 95% peak share in some regions
- **Search Engines:** Google dominates with 89% referral traffic, followed by Yandex (3.1%), Baidu (2.7%), Bing (2.6%)
- **Browsers:** Chrome remains the most popular browser globally
- **IPv6 Adoption:** Reached 28% of dual-stack traffic globally
- **Mobile Traffic:** 41% of global traffic came from mobile devices

# 225 Major Disruptions: Government Shutdowns Lead Outage Causes

TOTAL DISRUPTIONS

225

| CAUSE               | COUNT |
|---------------------|-------|
| Government Directed | 116+  |
| Cable Cut           | 41    |
| Unknown             | 16    |
| Power Outage        | 16    |
| Technical Problem   | 9     |
| Military Action     | 9     |
| Weather             | 8     |
| Maintenance         | 4     |
| Cyber Attack        | 4     |
| Natural Disaster    | 1     |

**Key Insight: Government-directed shutdowns account for over 50% of all major Internet disruptions in 2024**

The Cloudflare Radar Outage Center tracks scope and duration of these outages using Cloudflare traffic data

# 6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration



## Security Threats & Email Analysis

-  **Log4j Persistence:** CVE-2021-44228 continues to be actively targeted 2+ years after disclosure
-  **Malicious Emails:** 4.3% of all processed emails were malicious
-  **Deceptive Links:** 43% of malicious emails contained deceptive links to trick recipients
-  **Malicious TLD:** .bar originated the largest share of malicious and spam email
-  **RPKI Growth:** RPKI valid routes increased by 6.4% — improving Internet routing security

# Five Takeaways Shaping the 2025 Internet Outlook

1

## Traffic Growth Trajectory

17% global growth and 3.3x Starlink surge demonstrate resilient Internet demand expansion

2

## Post-Quantum Encryption Momentum

From 2% to 13% TLS 1.3 traffic — rapid adoption driven by Chrome 124

3

## Connectivity Fragility

225 disruptions with government shutdowns as leading cause highlights infrastructure vulnerability

4

## Persistent Security Threats

6.5% mitigation rate, Log4j persistence, and bot concentration demand continued vigilance

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## Evolving Developer Ecosystem

Go surpassed all languages to become the #1 API client language at ~12% of automated traffic